

Data Collection and Preprocessing Phase

Date	20 Sep 2026
Team ID	Master of Computer Applications
Project Title	Online Learning Engagement and Student Performance Analysis
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
online_education_dataset.csv	Missing Values in Platform Engagement Metrics: total_clicks, engagement level, and risk level contain 2,852 missing values each (~8.75% of dataset).	High	Impute missing total clicks with 0. Set missing engagement level to "None" / "Inactive" and map missing risk level to "Unevaluated" based on activity thresholds in Power Query.

online_ed ucation_ dataset.csv	Missing Performance Scores & Evaluations: engiscope has 5,866 missing values (~18.0%), and performance level has 5,898 missing values (~18.1%).	High	Replace missing engiscope with 0 or course-level median for unassessed students; set performance level to "Unassessed" / "Not Graded".
online_ed ucation_ dataset.csv	Missing Socioeconomic Data (sideband): 1,111 records (~3.41%) lack Index of Multiple Deprivation (sideband) classification.	Moderate	Impute null sideband entries with a distinct "Unknown" category string to preserve record count without biasing socioeconomic distribution.
online_ed ucation_ dataset.csv	Duplicate Student Records: 1,356 complete row duplicates and 3,808 repeated dissident	Moderate	Remove exact duplicate rows in Power Query; group or aggregate multi-attempt student IDs (dissident) by module attempt or retain latest attempt record.

	identifiers across course module attempts.		
online_education_dataset.csv	Inconsistent Categorical Formatting in sideband: Inconsistent string notation (e.g., "10-20" missing % symbol vs "20-30%").	Low	Standardize text patterns using Power Query string replacement ("10-20" → "10-20%") to ensure consistent categorical filtering and reporting.
online_education_dataset.csv	Incorrect Identifier Data Types: dissident is stored as an integer (int64), leading to auto-summing and numeric aggregation in analytical BI tools. Low	Low	Re-type dissident from Integer to Text/String data type to prevent accidental aggregation and ensure correct relational modeling.